

BIOL-8 Exam 03 — Answer Key

Modules 11-15: Skeletal, Muscular, Nervous, Microbiology & Cardiopulmonary Systems

Part A: Multiple Choice (31 points)

Question numbers: Module 11 items 1–6 = Q1–Q6; Module 12 = Q7–Q12; Module 13 = Q13–Q18; Module 14 = Q19–Q24; Module 15 = Q25–Q31.

Q#	Answer	Explanation
1	A	The skeletal system does not secrete digestive hormones.
2	B	Muscles contract and pull on bones; bones act as levers at joints.
3	C	Hematopoiesis occurs in red bone marrow.
4	D	The epiphyseal plate is hyaline cartilage where lengthwise growth occurs.
5	A	The femur is appendicular; the axial skeleton includes skull, vertebrae, ribs.
6	B	Synovial joints are freely movable.
7	C	Skeletal muscle moves the skeleton and generates heat.
8	D	Muscle → fascicle → muscle fiber → myofibril.
9	A	The sarcomere is the contractile unit between Z-lines.
10	B	Thick filaments are myosin.
11	C	Thin filaments slide relative to thick filaments; sarcomere shortens.
12	D	Acetylcholine is released at the neuromuscular junction.
13	A	CNS = brain + spinal cord.
14	B	PNS = nerves and ganglia outside the CNS.
15	C	Somatic division controls voluntary skeletal muscle.
16	D	The axon carries signals away from the cell body.
17	A	Myelin insulates and speeds saltatory conduction.
18	B	Reflexes can integrate in the spinal cord without conscious brain involvement for the initial response.

Q#	Answer	Explanation
19	C	Bacteria have no nucleus surrounded by a membrane; DNA is in the nucleoid.
20	D	Cocci, bacilli, and spirilla are the three common shapes.
21	A	Gram-positive cells retain crystal violet because of a thick cell wall and no outer membrane.
22	B	Binary fission is the usual bacterial division.
23	C	Survivors reproduce and pass resistance — natural selection.
24	D	Viruses need a host to copy themselves and do not carry out metabolism on their own.
25	A	Blood from the body returns to the right atrium first.
26	B	In the lungs, CO ₂ diffuses from blood into alveolar air to be exhaled; O ₂ moves from air into blood (wrong options claim O ₂ leaves blood into air or that you “feed” on inhaled CO ₂).
27	C	Arteries carry blood away from the heart.
28	D	“Lub” = AV (atrioventricular) valves closing between atria and ventricles (not semilunar).
29	A	Gas exchange happens in the alveoli .
30	B	Most oxygen is bound to hemoglobin in red blood cells.
31	C	Inhaling: diaphragm moves down and the chest volume increases .

Part B: Fill in the Blank (10 points)

Q#	Answer	Notes
1	periosteum	Outer fibrous covering of bone
2	axial	Skull, spine, ribs
3	appendicular	Limbs and girdles
4	actin	Thin filaments
5	troponin	Ca ²⁺ binding shifts tropomyosin
6	sympathetic	“Fight or flight”
7	dendrites	Receive incoming signals
8	peptidoglycan	Cell wall chemistry; Gram stain distinction
9	capsid	Protein coat around viral genome
10	acetylcholine	Neurotransmitter released by the motor neuron at the NMJ

Part C: Free Response (9 points)

Students choose 3 of 5 questions, 3 points each.

1. Bone cells

- **Osteoblasts** build bone (secrete matrix); **osteocytes** maintain bone in lacunae; **osteoclasts** resorb bone (1 pt)
- Accurate description of at least two cell types (1 pt)
- States why balance/remodeling matters for bone strength or calcium homeostasis (1 pt)

2. ATP and calcium

- **Calcium** exposes actin binding sites via troponin/tropomyosin (1 pt)
- **ATP** powers myosin detachment/cross-bridge cycling and relaxation (1 pt)
- If missing: e.g., no Ca^{2+} → no contraction; no ATP → rigor or cannot recycle cross-bridges (1 pt)

3. Autonomic divisions

- **Sympathetic**: fight-or-flight; **parasympathetic**: rest-and-digest (1 pt)
- One organ example for sympathetic (e.g., increases heart rate) (1 pt)
- One organ example for parasympathetic (e.g., promotes digestion) (1 pt)

4. Antibiotic resistance

- **Variation**: bacteria differ in sensitivity (1 pt)
- **Selection**: antibiotics kill susceptible cells; resistant survive (1 pt)
- **Reproduction**: survivors pass resistance to offspring (1 pt)

5. Cardiopulmonary integration

- Trace: right atrium → right ventricle → pulmonary arteries → lung capillaries (alveoli) → pulmonary veins → left atrium → left ventricle (2 pts if sequence correct; partial credit for minor gaps)
- Exercise link: e.g., higher cardiac output with increased ventilation to supply O_2 and remove CO_2 (1 pt)

End of Answer Key